

Assembly Test Stand: Construction of the Barrel Timing Layer for CERN

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Abstract

This project involves constructing an Assembly Test Stand to help with the proper installation of the scintillator silicon photomultipliers (Scintillator SiPMs) to a tray that will be installed into the Barrel Timing Layer of the Compact Muon Solenoid in the Large Hadron Collider at CERN. I have referenced the CAD file of the Assembly Test Stand provided by engineers at CERN and used it to order the necessary parts for its construction from McMaster and manufacture parts via metal working equipment. I built the Assembly Test Stand from these parts and modified parts as necessary for the build. During the assembly process, there was a lack of communication with the design, so I consulted with the engineers at CERN who created it. I have also manufactured parts incorrectly, therefore, I noted my mistakes and bought new stock to recreate the parts correctly. Even with these difficulties, the overall construction of the test stand was a success. The Assembly Test Stand successfully holds the tray, but the installation of the Scintillator SiPMs has not started, so its success is still undetermined. The next steps are to work on the installation and improve the Assembly Test Stand as needed. \

Background

The Large Hadron Collider at CERN in Switzerland Geneva was constructed to carry out high-energy particle collisions between hadrons and observe the results of these collisions to better understand subatomic phenomena and the standard model of particle physics. As CERN discovers more about the nature of subatomic particles, such as discovering the Higgs-Boson, the detectors at the Large Hadron Collider need to be updated to continue this journey of discovery. One of the updates is called the High Luminosity Large Hadron Collider Phase, where each of the detectors in the particle collider is updated to handle collecting data from more particle

collisions than before.¹ The more particles collide, the higher the chances there are of new/rare subatomic particles occurring from these collisions.

Within the Large Hadron Collider, there are four collision points. One of those collision points is at the Compact Muon Solenoid (CMS) which is a “general-purpose detector; that is designed to observe any new physics phenomena that the LHC might reveal.”² The Minimum ionizing particle timing detector (MTD) will be installed in the CMS to detect the timeline of events more precisely regarding the minimum ionizing particles within the particle collisions that take place. In more detail,

“The precision time information from this MIP Timing Detector (MTD) will reduce the effects of the high levels of pileup expected at the HL-LHC, bringing new capabilities to the CMS detector”³

Within the MTD is the Barrel Timing Layer (BTL) which detects the timing of individual photons entering/leaving the collisions which are used to determine the timing of minimum ionizing particles in the collisions through the MTD. The Barrel Timing Layer is a thin, cylindrical detector that is made up of trays that hold many silicon photomultipliers that allow it to perform the task at hand.³

Caltech is one of four sites around the world that is responsible for manufacturing one of these trays, installing silicon photomultipliers onto the tray, and shipping the tray to CERN to build the BTL. To properly install the photomultipliers onto the tray, a tray assembly stand needs to be manufactured. The tray assembly stand allows the tray to be rotated and set to various rotational degrees to more properly install and secure the delicate photomultipliers onto the tray. It also allows both sides of the tray to be accessed at the same time, which is crucial for installation. Properly installing these photomultipliers onto the tray is important, as once the BTL

is installed within the MTD, and therefore installed within the CMS detector, it may not be removed until the next update phase of the particle collider (which will be years later).

Results

The construction of the Assembly Test Stand is complete, and the tray has been installed onto the Assembly Test Stand. It is fully functional, as it can rotate the tray successfully with ease at any 45-degree angle. The test stand also allows the tray to be oriented in any manner.



Figure 1: The picture above shows the complete build of the Assembly Test Stand with one of the trays attached to it in its upright position.



Figure 2: The picture above demonstrates the rotational ability of the top section of the Assembly Test Stand. The top section is rotated at a 90-degree angle.

The stand has not been used for the installation of Scintillator Silicon Photomultipliers or used for testing the tray's functionality. It is unknown whether the test stand can properly help with the execution of testing the tray because of this. The cooling tubes have not yet been

installed into the tray, yet the Acetal plastic attachments that interact with the tray itself to secure it to the test stand have a low thermal conductivity at $.86 \text{ Wm/K}$.⁴ The 3D printed PLA spacers that are placed between the acetal plastic and the tray have a low thermal conductivity of 0.183 Wm/K ⁵ and the M2.5 screws have a thermal conductivity of 16.3 Wm/K ⁶. While this value is higher than ideal, it has the lowest thermal conductivity of all the metals. Material with low thermal conductivity interacting with the tray is important for the cooling tubes as it allows the tray to cool more efficiently.

The results of this SURF project are successful; the test stand is constructed with the tray attached. It properly supports the tray as well as all its electronic components that detect individual photons (Scintillator SiPMs) and transmit this data to the CMS computer servers at CERN. This allows Caltech to successfully assemble the scintillator SiPMs onto each of the 18 trays they must construct for the BTL at CERN in Spiropulu's lab. Onward, the design for the assembly test stand will be updated based on the needs of Spiropulu's lab while assembling the electronic components of each of the trays. When the assembly test stand design is refined, the build of the test stand will be reconstructed as necessary to reflect the design changes to help with a more efficient installation of the Scintillator SiPMs and other electronic components onto the trays.

Methods

A STEP Assembly file from one of the research design engineers at CERN was received. The project goal was to construct the assembly test stand at Caltech with this file. Below is an outline of how this task was completed.

Determined what was custom-made and what was a standard part of the design.

A bill of materials for the Assembly Test Stand was created to keep track of what materials were necessary to purchase for its construction. Each part of the stand in the STEP file was highlighted and searched for on the McMaster-Carr website. If the part was found on the website, then it was a standard part that could be bought and shipped to Caltech. Otherwise, the correct-sized stock would be bought for its custom manufacturing job and shipped to Caltech from McMaster-Carr. All measurements for each part were taken using the measurement tool available in SolidWorks (the CAD software used to access the STEP file). Once the order of materials/parts was finalized on the McMaster-Carr website, then it was used as a reference to complete the Bill of Materials for the Assembly Test Stand. The BoM was then sent to one of the research project engineers for revision and approval to make sure the materials accurately reflected the makeup of the design.

Ordered all the stock/parts I would need for the assembly

The Bill of Materials (BoM) was revised several times. Once the BoM was finalized, all the material was ordered from McMaster-Carr. More small orders were made after this order because of mistakes that were made in the manufacturing process, or to reflect changes made to the design of the Assembly Test Stand during its construction. Extra material from construction was collected into a bin as scrap metal or brought to a recycling center.

Created drawing files for custom-made parts

Once all the material had been shipped to Caltech from McMaster-Carr, drawing files for the custom-made parts of the design needed to be created. Each custom part was manufactured from the STEP file and exported separately from the rest of the file by using the “pack and go” feature in SolidWorks of the part. These individual part files were opened, and for each one, the

“Make Drawing from Part” button was selected. This would open a drawing file of the part where one could select the necessary format of the drawing file as well as its necessary drawing views. Each part was then properly dimensioned in metric units and annotated with notes on necessary information about the manufacturing process of the parts. The drawing files were shown to Jason Trevor, the co-mentor, for revision and finalization.

Outsourced difficult jobs to the GALCIT lab

There was a total of eight custom parts or four custom parts that each had a duplicate. Half of these parts were outsourced to the GALCIT lab for manufacturing (these were the rotational shafts and the circular plates), and the other parts were manufactured. Added to the outsourced job was one more custom part that required 24 copies, which required a CNC milling machine. For each of the custom parts, the drawing files were sent to the GALCIT lab as well as the individual part files of the custom parts all through email. Any issues or questions the GALCIT lab had regarding these jobs would be answered through email.

Manufactured by hand the other custom parts

The other custom parts were manufactured by hand. A vertical milling machine in the Jim Hall Design and Prototyping Laboratory at Caltech was used to create these parts. This lab allowed student access, which was why it was used. These rectangular parts were each measured with a ruler and cut down to size using a band saw. Then the vertical milling machine was used for milling the size of each part to accuracy, slotting, drilling, and reaming. Any holes that needed to be tapped were hand-tapped. Zeroing the part each time it is moved within the vise is a critical step for accuracy, and led to manufacturing mistakes when these parts were made for Caltech’s Assembly Test Stand. The twenty-four spacers used for the twenty-four tray attachments were 3D printed.

Used a bottom-to-top approach in the assembly

Once the custom parts were outsourced/handmade, the Assembly Test Stand was constructed. The stand was constructed via a bottom-to-top approach. The bottom frame was constructed first, then the middle frame was constructed on top of the bottom frame, the custom-made attachments for the top section were attached to the middle frame, then the top frame was built separately and attached to the rest of the test stand.

Firstly, each of the standard parts was cut down to size either using a band saw or a rotary saw. The three 10 ft 40 x 40 mm t-slot framing rails were cut down to 275 cm (111.81 in), the stems on all six casters were cut to roughly 10 mm, and the two 10 ft 45 x 45 mm t-slot framing rails were cut to 275 cm (108.27 in). Secondly, the bottom frame was assembled with two 275 cm framing rails, two 61 cm (24 in) framing rails, and one 30.5 cm (12 in) framing rail via 40 x 40 mm single closed gusset t-slot corner brackets. All six casters were attached to the underside using alignment nuts in the t-slots of the framing rails. Thirdly, the middle section was built extending from the bottom section with one 275 cm framing rail, two 91.4 cm (36 in) framing rails, four 30.5 cm (12 in) framing rails, and one 45.7 cm (18 in) using the same gusset t-slot corner brackets as well as diagonal single gusset diagonal brackets. The four 30.5 cm framing rails were threaded with an M8 hand tap on both sides for the usage of diagonal brackets. The top-facing sides of the two 91.4 cm framing rails were also tapped with an M8 hand tap.

Fourthly, the custom-made attachments for the top section were attached to the middle section by attaching the base plates using an M8 screw and 5 mm alignment dowel pins. Then separately, the attaching plate to the top section, the rotational shaft through the bearing, and the circular plate were attached via a press fit. These two assemblies were attached to each base plate via M10 screws. Fifthly, the top section was built separately with two 30.5 cm (1 ft) 45 x 45 mm

t-slot framing rails and two 275 cm 45 by 45 mm t-slot framing rails via t-slot corner brackets. Lastly, the two 275 cm framing rails were threaded using an M12 tap on both sides with M12 screws screwed partially into the framing rails. The top section is then attached to the rest of the assembly stand by sliding the M12 screws attached to the top section into the slots of the attachment plates for each side, then tightening them.

To attach the tray to the test stand, twenty-four acetal plastic attachments were attached to the test stand using two M8 screws and two M8 t-slot alignment nuts for each plastic attachment. Then, the attachments as well as the attachment spacers were screwed into the tray using M2.5 screws and optionally M2.5 nuts.



Figure 3: To the left is a close view of the tray attached to the test stand. The black attachments are screwed into the test stand and screwed into the tray to attach the two together.

Consulted project engineers to discuss holes in the design

During construction, I realized that some aspects of the design sent by CERN were not made clear. The main components that did not appear in the STEP file for the design were the fasteners/attachments for the Assembly Test Stand. To solve this, I would consult the project engineers who sent me this design through email or Zoom. The methods of attachment/fastening that were unclear have been documented in the drawing files of custom parts and in the Bill of Materials for this project. There were some parts removed from the design because of a lack of

attachments in the design. These parts were the double t-slotted framing rails that would attach under the top section of the Assembly Test stand for extra support. After a discussion with the project engineers, we determined that these were not necessary to install for Caltech's prototype of the Assembly Test Stand.

Changes were made to the original design

Related to the holes in the design, the changes made to the test stand design affected the fasteners and attachments used. The fastener between the rotational shaft and the attachment plate for the top section of the test stand was changed from a screw to a pin.



Figure 4: The above shows the screw fastener to attach the rotational shaft and attachment plate for the top section of the test stand.



Figure 5: The above shows the pin fastener to attach the rotational shaft and attachment plate for the top section of the test stand.

Another change was made to the custom tray attachments by making them a uniform thickness, whereas before this was not the case as shown below:

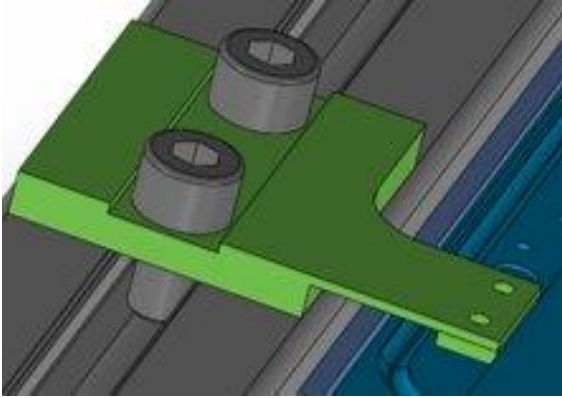


Figure 6: This picture shows the original design of the tray attachment.

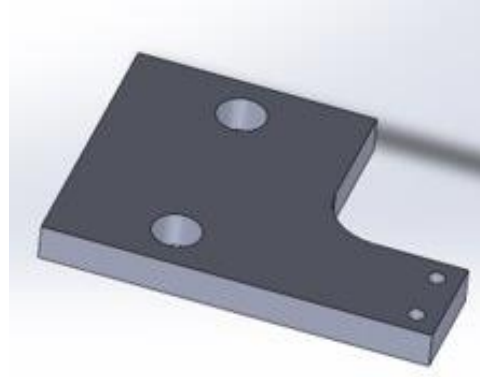


Figure 7: This picture shows the modified design of the tray attachment.

These changes to the design were both easy to fix and implement in the current prototype at Caltech.

Acknowledgments

I would like to express my deepest gratitude to my co-mentor and assistant research project engineer Jason Michael Trevor for passing on his invaluable knowledge about design principles and construction in the engineering field. We worked together to solve mistakes of mine and he taught me various manufacturing skills that can be applied to the project and future use. He taught me how to outsource custom parts and refined my skills in creating drawing files/CAD files. He also helped me with cutting large t-slot framing rails that were critical for the stability of the Assembly Test Stand. I would not have completed this project if it were not for him.

I am extremely grateful to Maria Spiropulu for guiding my project when it became stuck. She taught me how to communicate with my coworkers and the engineers at CERN about issues I ran into while completing the build. She showed me that I could work with the engineers back

at CERN to conclude changes to design flaws instead of assuming that I must have interpreted something in the design incorrectly. She also taught me perseverance in communication, an invaluable skill.

I could not have completed this project without the Carl F. Braun trustee to fund half of the payment for my SURF (Summer Undergraduate Research Fellowship) with Spiropulu's Lab. Without them, I would not have been granted a SURF for the summer of 2023. This project was truly an invaluable experience to work on, as I have learned so much about engineering and communication. I got to pursue my dream of working for CERN because of them.

I would like to extend my thanks to Adi Bornheim and Guillermo Reales for working with me on the design for the Assembly Test Stand and collaborating with me on design changes/clarifications for the stand. I also appreciate their kindness and patience with me. I would like to thank the GALCIT lab for their work on creating the outsourced parts for this project, as well as their patience and kindness when teaching me about engineering and manufacturing principles.

Lastly, I would like to thank my coworkers Kai Svenson and Sherry Wang as well as my family. These people brought plenty of moral support and encouraged me to not give up when I ran into roadblocks with my project.

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Appendices

Appendix A: Drawing Files of Custom Parts

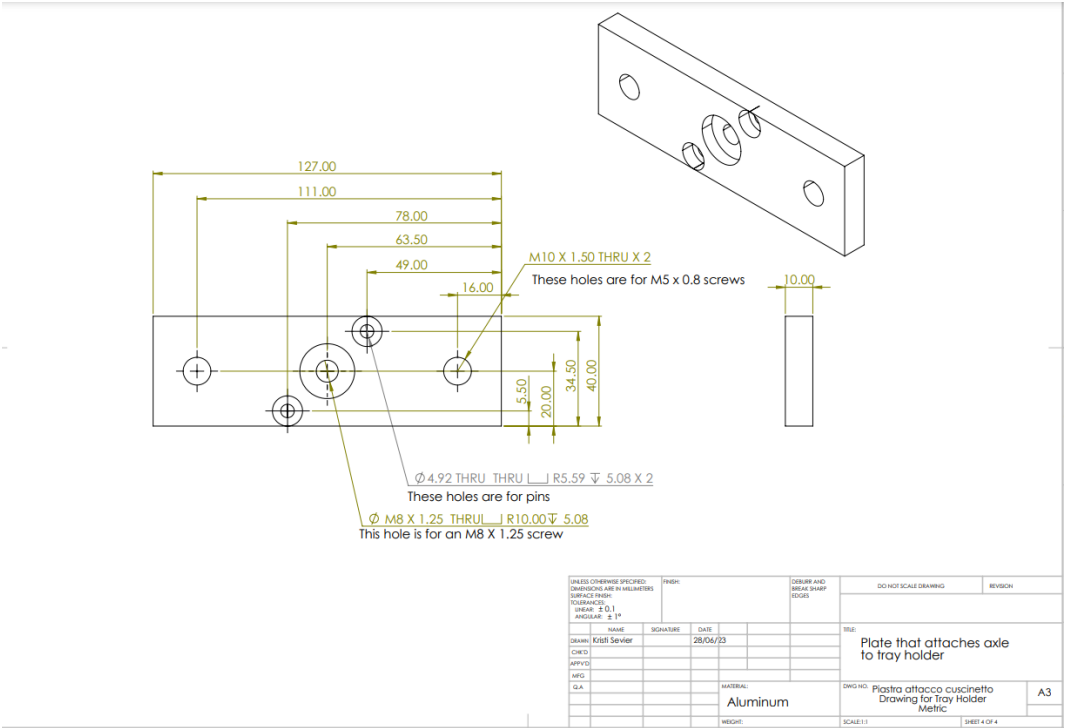


Figure A1: Base Plate for Rotational Assembly of Test Stand

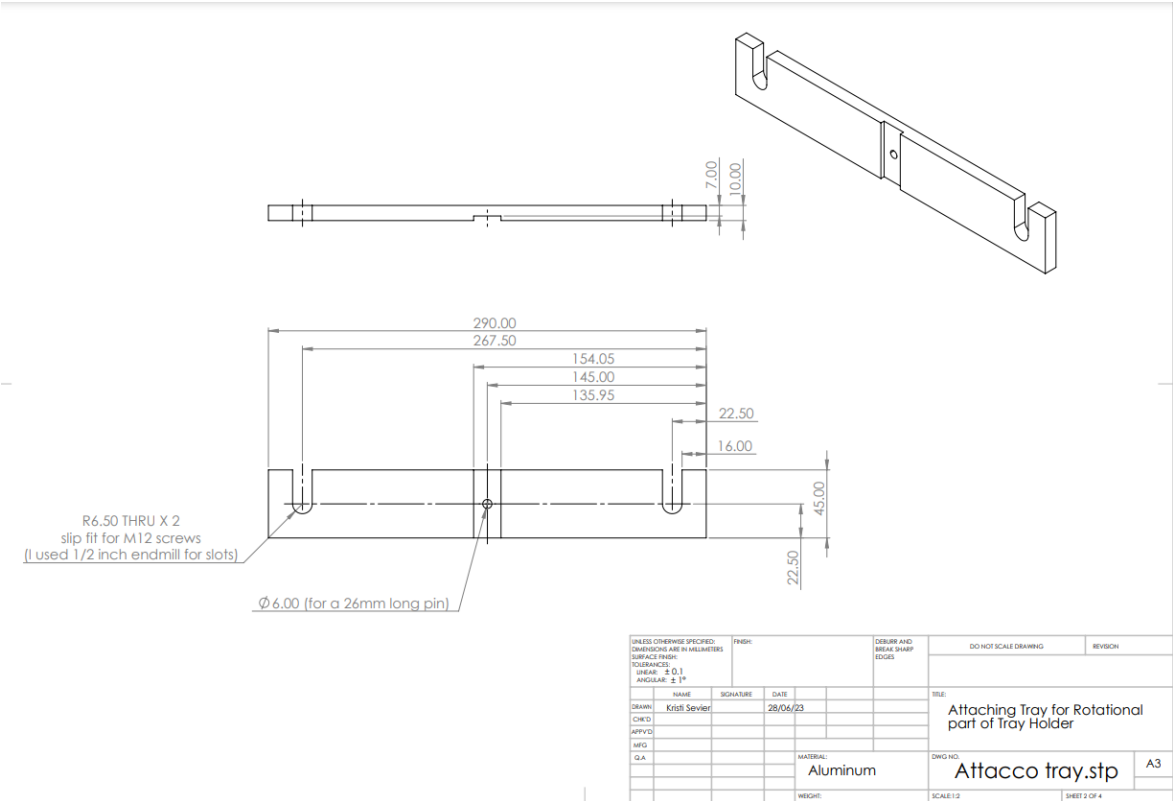


Figure A2: Attachment Plate for the Top Section and the Rotational Assembly of Test Stand

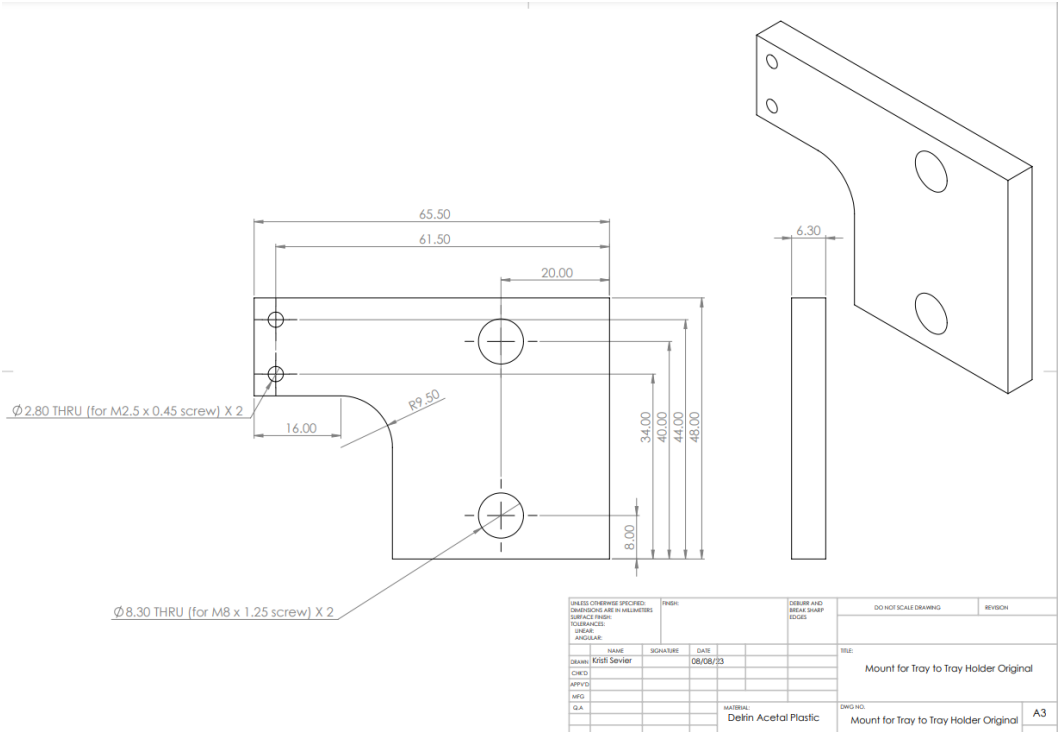


Figure A3: Tray Attachment (or mount) for the Test Stand

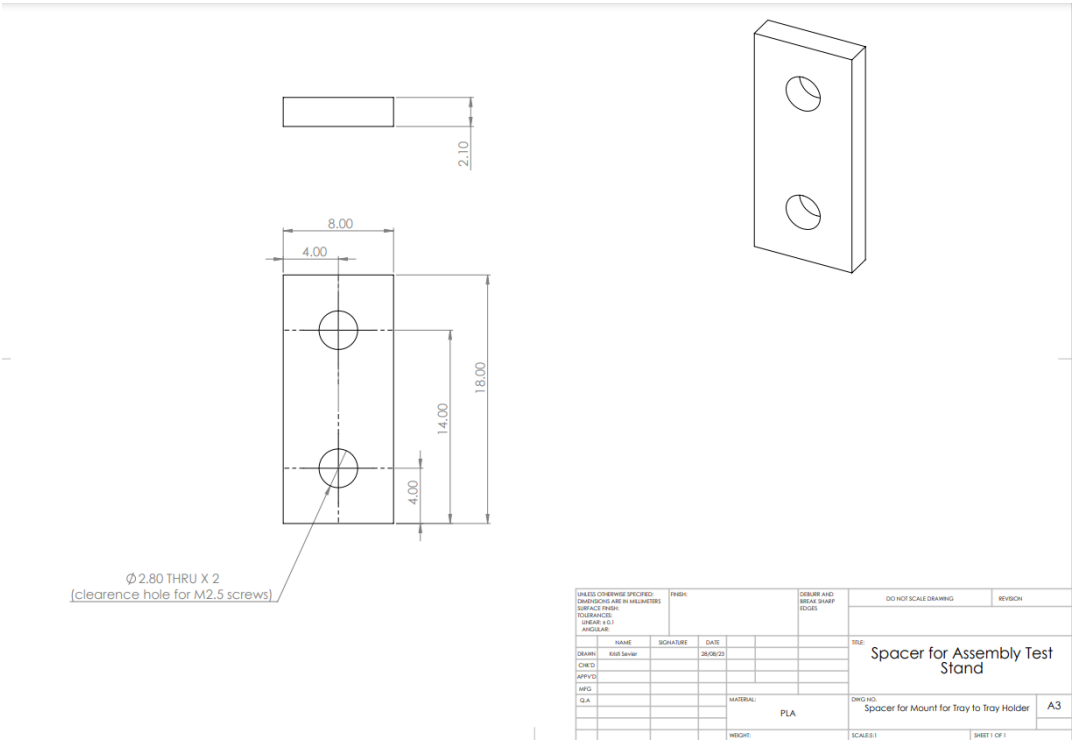


Figure A4: Spacer for Tray Attachment (or mount) for the Test Stand

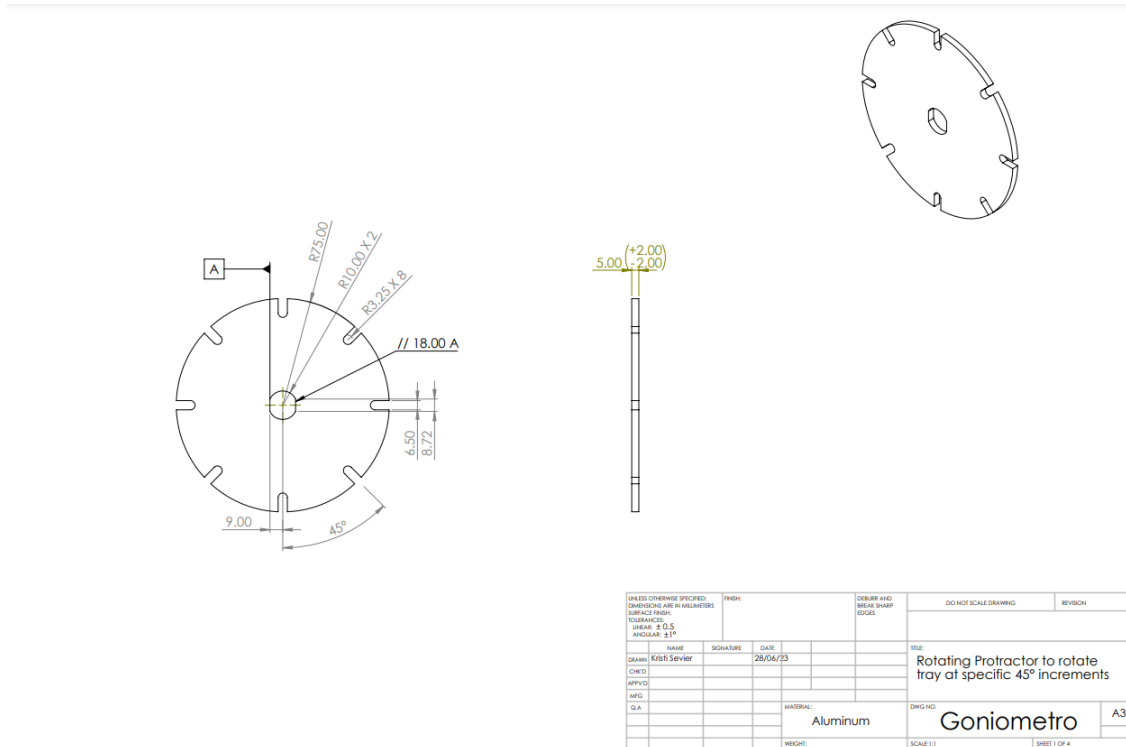


Figure A5: Circular plate for Rotational Assembly of Test Stand

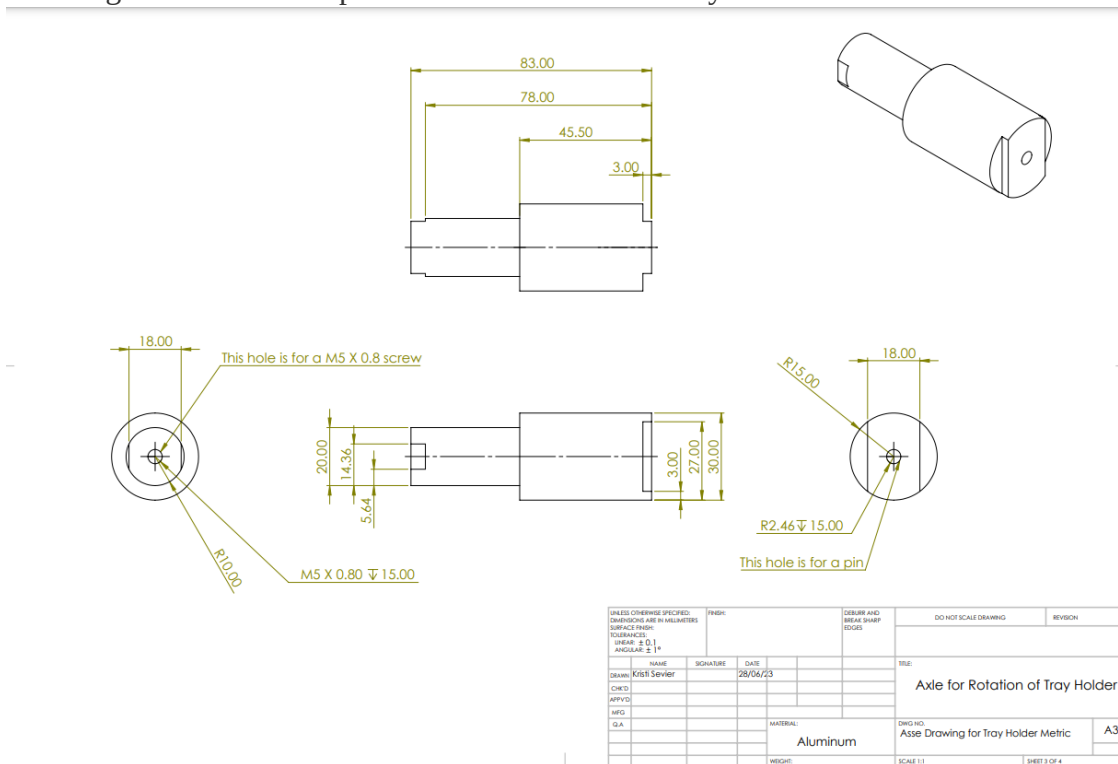


Figure A6: Shaft for Rotational Assembly of Test Stand

Appendix B

Link to the Bill of Materials for this project:

[Final Bill of Materials for Assembly Test Stand](#)